

GSM-Based Borewell Water-Level Monitor

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If the water level in a borewell drops below the threshold level for pumping, its pump motor may get air-locked or even burn out due to dry running. It is inconvenient for

farmers to walk all the way to their fields at night just to switch the pump motor 'off.' Besides, he may never get to know the problem.

This problem can be solved by using this GSM-based system that will automatically give the user a

call or send an SMS from his mobile phone when the water level in the borewell drops below or rises to the threshold level for pumping. The user can also remotely switch on or switch off the pump motor

Circuit description

Fig. 1 shows the block diagram of the GSM-based borewell water-level monitoring system. Fig. 2 shows the complete circuit. It comprises the power supply section, water-level sensor circuit, microcontroller, MAX232 driver, relay driver and GSM modem. The GSM board has a valid SIM card with sufficient recharge amount to make outgoing calls.

The circuit is powered by regulated 5V DC. The 220V, 50Hz AC mains is stepped down by transformer X1 to

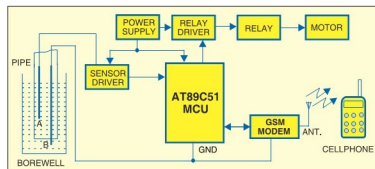


Fig. 1: Block diagram of GSM-based borewell water-level monitoring system

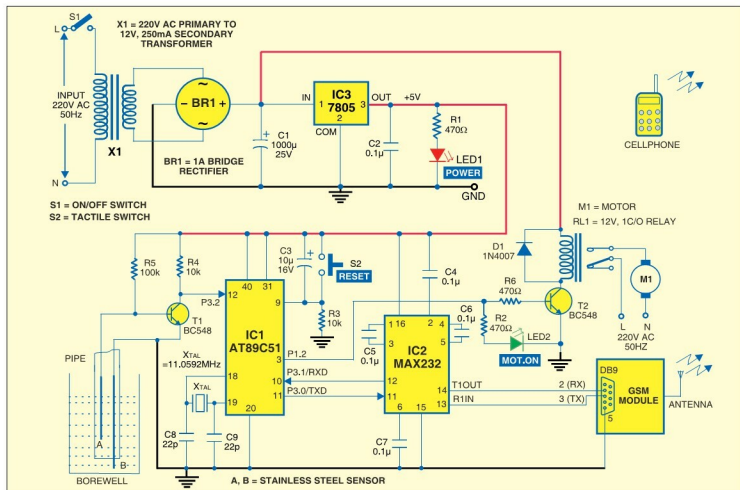


Fig. 2: Circuit of GSM-based borewell water-level monitoring system